Data Analysis Section

Summary of the Selected Dataset

The Korean Tax Avoidance Panel (KoTaP) is a long-term panel dataset. This dataset contains 12,653 firm-year observations from 1,754 firms between the years 2011 and 2024. The dataset consists of firms that are listed on the KOSPI and KOSDAQ sock exchanges in Korea.

The dataset includes 65 variables related to profitability, stability, growth and governance of the firms as well as common measures of tax avoidance. The data has been compiled by academics in Korea specifically for use in econometrics and machine learning projects related to tax avoidance.

Tax avoidance is measured through two categories of indicators.

- Effective tax rates:
    - CETR: The cash effective tax rate is the amount of cash taxes paid divided by pre-tax income.
    - GETR: The GAAP effective tax rate is the total tax expenses as recognized by GAAP accounting standard divided by pre-tax income.
    - The dataset also includes three and five year averages for these two indicators, as well as an “adjusted” version that shows the average tax rate relative to comparable firms in the same industry and of similar size.
- Book-Tax difference indicators
    - TSTA: The total book-tax difference is the difference between accounting income and taxable income on an accrual basis
    - TSDA: The discretionary book-tax difference is the difference between accounting income and taxable income based on discretionary accruals.

The dataset also includes a range of financial indicators. These include:

- SIZE: a measure of the scale of the firm, measured as the natural logarithm of total assets
- LEV: a measure of leverage, calculated as total liabilities divided by total assets
- ROA: Return on assets, calculated as the net income divided by the lagged total assets
- CFO: Operating cash flow scaled by total assets
- MB: Market to book ratio, calculated as the market capitalization divided by the book equity

- TQ: Tobin's Q, which is the market capitalization plus total liabilities, all divided by total assets.

The dataset also includes a range of indicators for governance characteristic, including:

- BIG4: an indicator that they are audited by a big 4 accounting firm
- FORN: the percentage of foreign ownership
- OWN: largest shareholder ownership percentage
- KOSPI: a dummy to indicate which market they are listed on

Descriptive Statistics

The general coverage and composition of the dataset is detailed in the table below.

Panel Coverage and Composition	
Firm-Year Observations	12,653
Unique Firms	1,754
Years Covered	2011-2024
Proportion of Observations by Stock Exchange	KOSPI 44%, KOSDAQ 56%

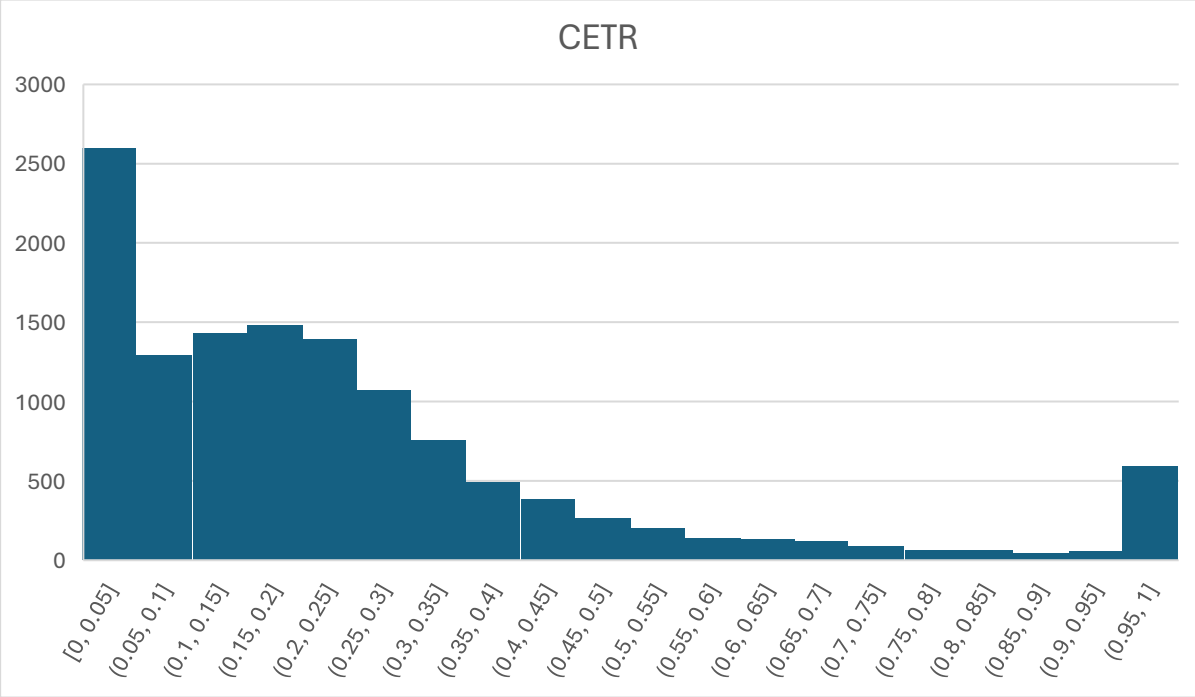
Key tax avoidance features are detailed in the table below. The authors have cleaned the dataset to ensure there are no missing values, so the number of observations for all of theses features is 12,653.

Key Features					
Feature	**Description**	**Range**	**Mean**	**Median**	**StdDev**
CETR	Cash Effective Tax Rate	0.000 to 1.000	0.244	0.184	0.245
GETR	GAAP Effective Tax Rate	0.000 to 1.000	0.202	0.201	0.154
TSTA	Total Book-Tax Difference	-3.328 to 0.528	−0.0847	-0.019	0.286
TSDA	Discretionary Book-Tax Difference	-3.427 to 0.526	−0.0931	-0.026	0.285

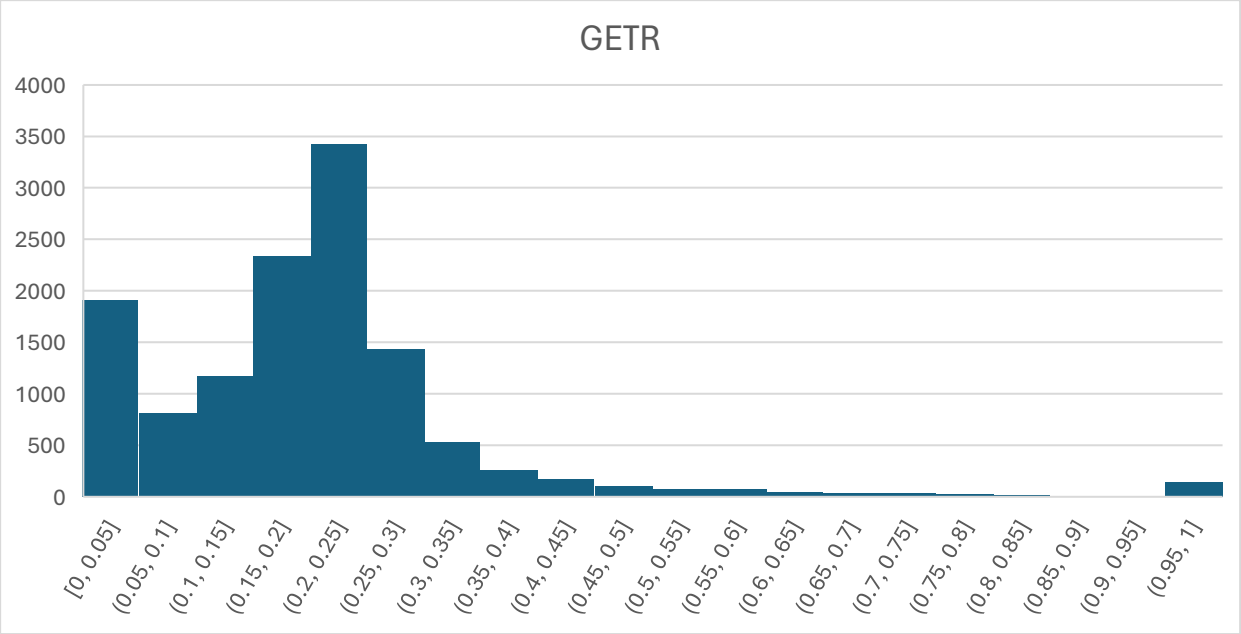
CETR3	Three-year average CETR	0.000 to 1.000	0.218	0.185	0.201
GETR3	Three-year average GETR	0.000 to 1.000	0.202	0.204	0.150
CETR5	Five-year average CETR	0.000 to 1.000	0.200	0.189	0.145
GETR5	Five-year average GETR	0.000 to 1.000	0.200	0.206	0.125
A_CETR	Adjusted Cash Effective Tax Rate	-0.245 to 0.824	0.053	-0.010	0.240
A_GETR	Adjusted GAAP Effective Tax Rate	-0.236 to 0.764	0.004	-0.003	0.147
A_CETR3	Adjusted three-year average CETR	-0.235 to 0.815	0.019	-0.017	0.195
A_GETR3	Adjusted three-year average GETR	-0.245 to 0.782	.001	-0.005	0.143
A_CETR5	Adjusted five-year average CETR	-0.264 to 0.771	-0.031	-0.044	0.139
A_GETR5	Adjusted five-year average GETR	-0.255 to 0.754	-0.013	-0.014	0.118
SIZE	Natural logarithm of total assets	23.621 to 30.622	26.245	25.996	1.348
LEV	Leverage = Total Liabilities / Total Assets	0.026 to 0.871	0.325	0.313	0.177
ROA	Return on Assets = Net Income / Lagged Total Assets	-0.230 to 0.295	0.063	0.048	0.056
CFO	Operating Cash Flow scaled by total assets	-0.279 to 0.311	0.072	0.064	0.079
MB	Market-to-Book Ratio = Market Capitalization / Book Equity	0.235 to 13.192	1.490	1.053	1.456

TQ	Tobin's Q = (Market Capitalization + Total Liabilities) / Total Assets	0.415 to 8.095	1.323	1.034	0.973
BIG4	Big4 audit dummy	0.000 to 1.000	0.489	0.000	0.500
FORN	Foreign ownership share (%)	0.000 to 1.000	0.085	0.035	0.120
OWN	Largest shareholder ownership share (%)	0.023 to 1.000	0.437	0.432	0.153

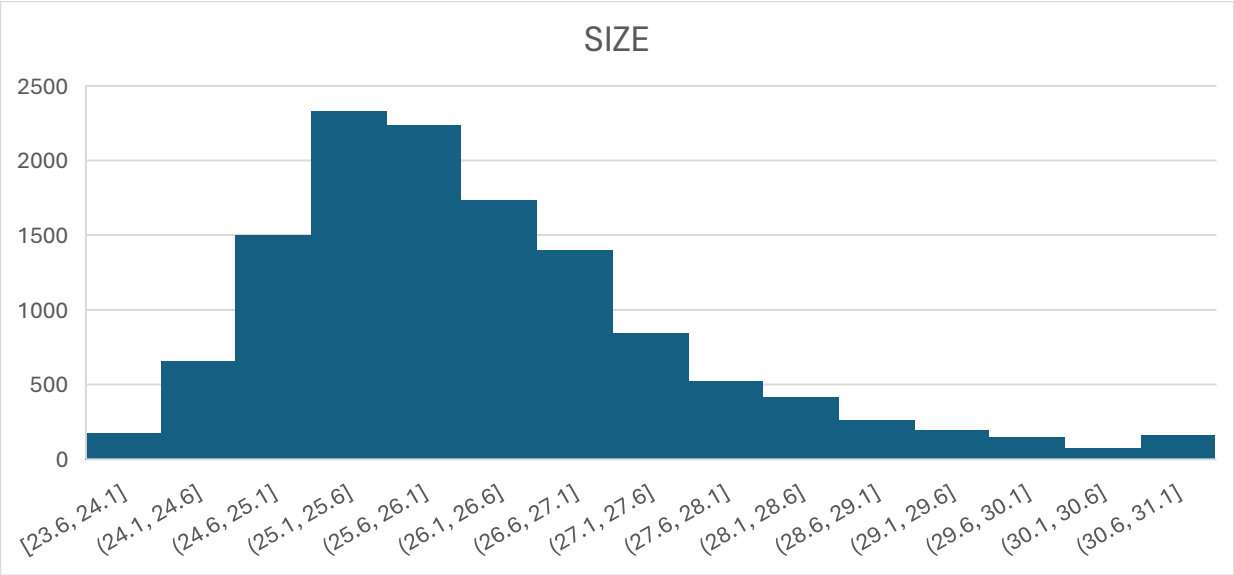
Looking at a histogram of CETR, we see a dense cluster of values between approximately 5% and 25% and a large number at or near zero. There is also a small spike in values at or near 100%.



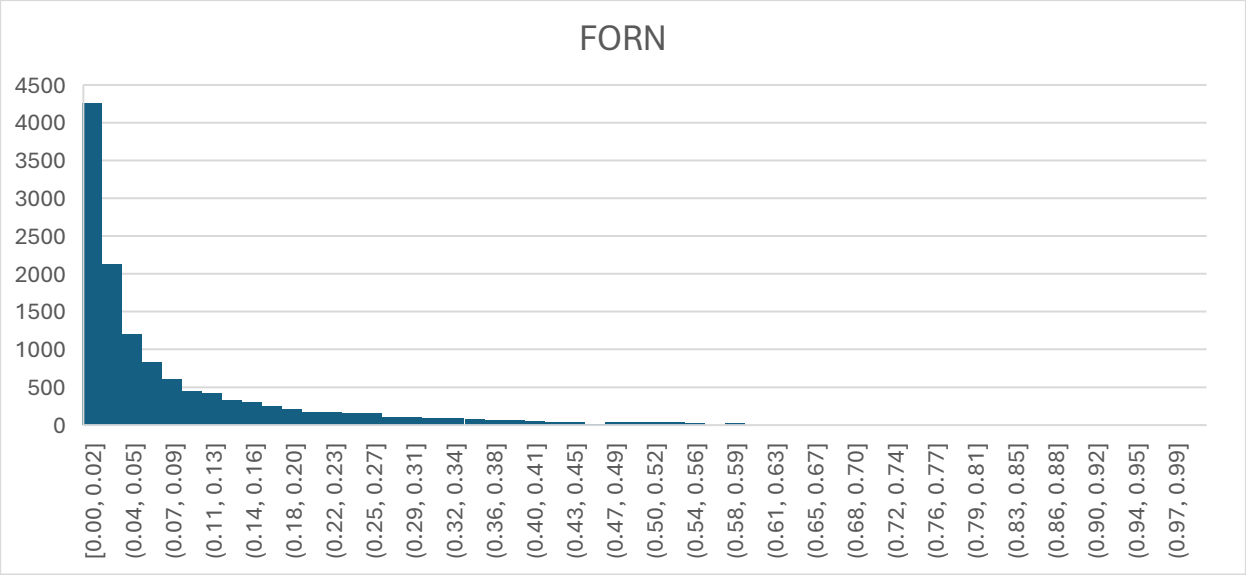
A histogram of GETR follows roughly the same pattern but with a more pronounced concentration of values between about 15% and 25% and fewer values near 0 or 1. This is reflective of the GAAP accounting rules.



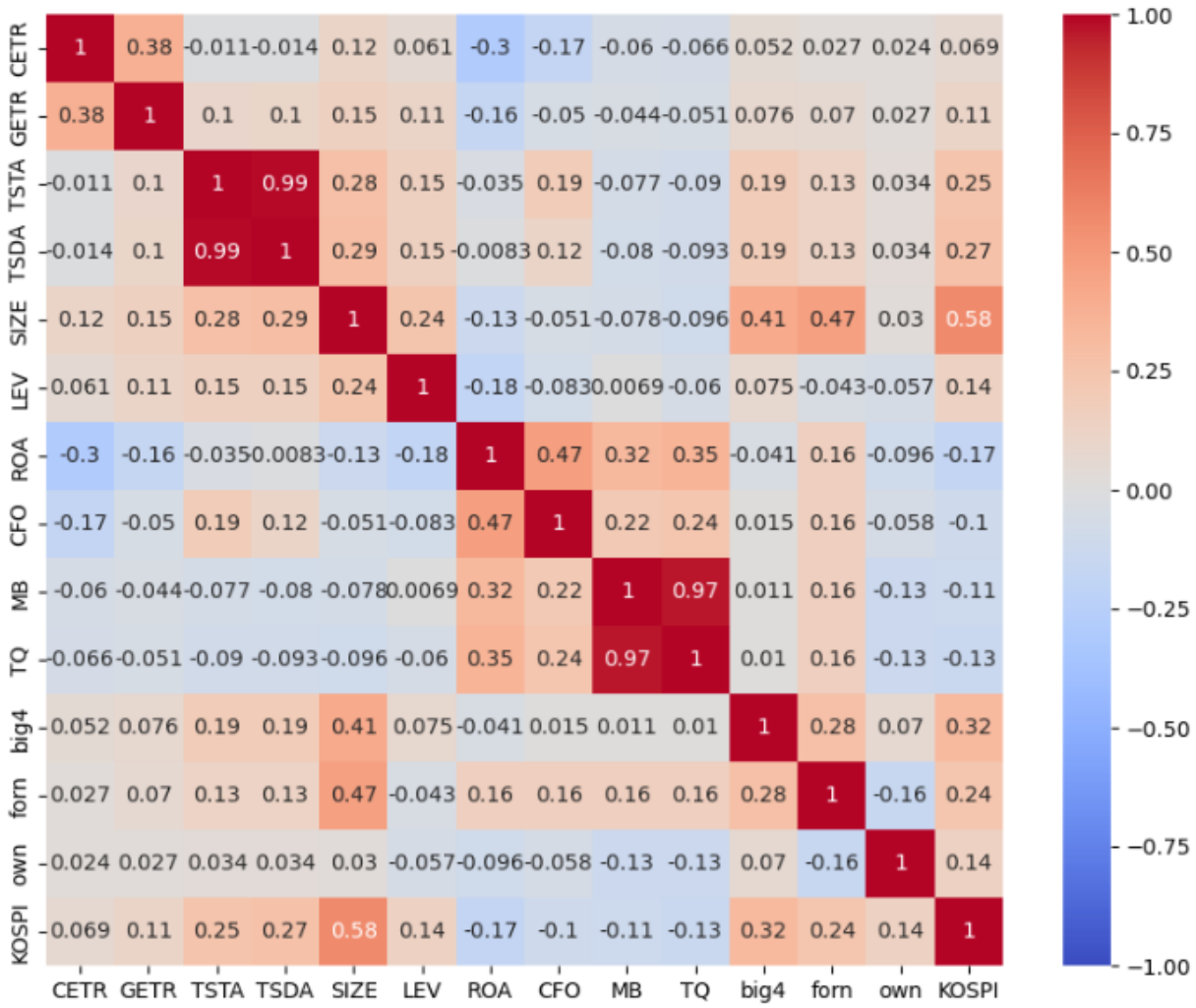
A histogram of SIZE shows a skewed bell curve shape, with a concentration of values between approximately 25 and 27. This corresponds to a concentration of mid sized firms in the dataset, with long upper tail indicating a subset of very large firms in the dataset. There are relatively few small firms.



A histogram of FORN shows a high concentration near zero, indicating that most firms have very little foreign ownership. Because it is skewed it may need to be standardized or normalized for use in some models.



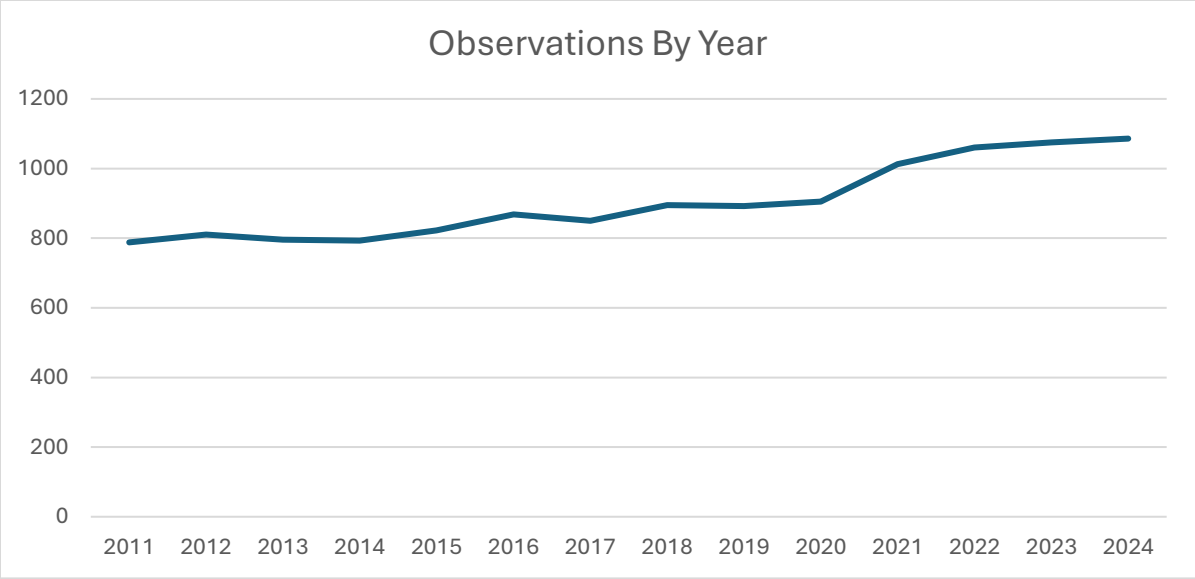
Looking at a correlation matrix, we see very strong correlation between TSTA and TSDA, which is expected as they capture closely related concepts. CETR and GETR also show a moderate correlation, which is expected as they also measure related concepts, but shows the greater difference between cash based and GAAP based effective tax rates. CFO and RAO also show moderately strong correlation, reflecting the link between profitability and operating cash flow. MB and TQ also correlate strongly, as they are both market based assessments of firm value. In general, aside from some extreme multicollinearity between features that are measuring very similar things, most cross domain correlations are moderate or low, suggesting that they provide unique information that could make them useful in this analysis.



Constraints and Data-Quality Notes

The researchers that compiled this dataset did extensive work to clean and prepare the dataset for analysis, so there are very few limitations.

One important note is that the number of firms is not the same for each year, so the panel isn't balanced. As shown in the chart below, the number of observations increases by year. To account for this, I plan to ensure that the train/test splits are done by year, not randomly, to respect the temporal structure of the dataset.


Observations By Year
1200
1000
800
600
400
200
0
2011
2012
2013
2014
2015
2016
2017
2018
2019
2020
2021
2022
2023
2024